Université Nice Sophia Antipolis

Master 2 Thesis

Recovery of wave profiles through pressure measurements at the bed

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I would like to thank my thesis advisor M. D. Clamond for introducing me to his original works regarding wave profile recovery methods. More importantly, I would like to thank him for his great patience and valuable guidance.

I, Jad Tabbouch, declare that this thesis titled, '*Recovery of wave profiles through pressure measurements at the bed*' and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a research degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
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Abstract

The aim of this master's thesis is to present two methods of recovering the sea wave profiles via pressure measurements at the bottom of the sea bed.

To this aim - in Chapter 1 - we present and justify the physical hypotheses of the chosen model, presenting the basic physical relations which will be used throughout the paper.

Chapter 2 will provide the main results in analysis and harmonic analysis to prepare for the formulation of the problem in terms of Stokes waves presented in Chapter 3. Subsequently, we derive useful properties and inequalities for the pressure and velocity which will be necessary for the two methods of surface recovery presented.

Chapter 4 deals with the first method of recovery of the surface elevation as presented by D. Clamond and A. Constantin's paper 'Recovery of steady periodic wave profiles from pressure measurements at the bed' (2012), in the form of implicit integral equations.

A simpler, more effective way of recovering the surface elevation will be discussed in Chapter 5, as appears in D. Clamond's 'New exact relations for easy recovery of steady wave profiles from bottom pressure measurements' (2013).

The results obtained in the latter allow for an algorithmic approach to effectively compute the surface elevation with experimental data obtained directly by provided pressure measurements in Chapter 6.

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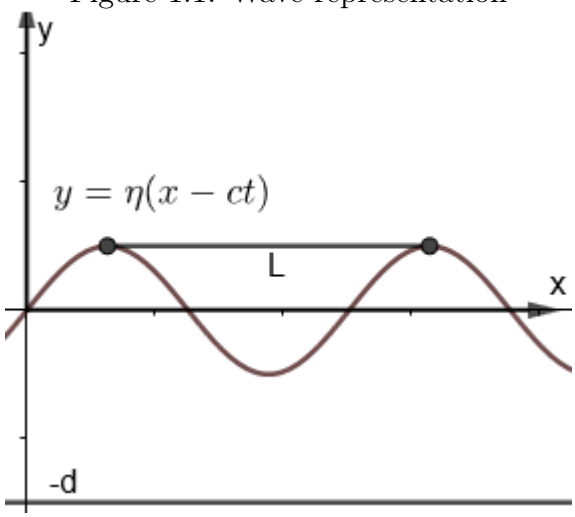
Chapter 1

Model description and physical hypotheses

Let us consider a two-dimensional periodic wave, for which the motion is identical in any direction parallel to the crest line and periodic in the spatial direction in which the wave is propagating.

To simplify, consider a cross section of the flow that is perpendicular to the crest line with coordinates (x, y), the $y-$axis pointing vertically upwards, and the $x-$axis being the direction of wave propagation, taking the origin as $y = 0$.

Figure 1.1: Wave representation



Let $(u(t, x, y), v(t, x, y))$ be the velocity field of the flow over the flat bed at $y = -d$, where $d > 0$ is the depth of the water. In Chapter 3, we shall

make it clear how we eliminate the time dependency by passing to a moving frame (in the context of Stokes waves). We will now suppose that it has been done in order to simplify notations.

Let $y = \eta(t, x)$ be the water's free surface. Since $y = 0$ is the mean water level, we have for all t,

$$\int_0^L \eta(t, x)dx = 0 \tag{1.1}$$

where $L > 0$ denotes the wave period. The functions η as well as u and v are $L-$periodic in the x-variable. For any $2\pi/k$ periodic wave profiles we shall denote from now on the Eulerian average operator by $\rangle, \langle$ (average taken over one wavelength):

$$\langle \eta \rangle = \frac{k}{2\pi} \int_{-\pi/k}^{\pi/k} \eta(x)dx \tag{1.2}$$

We define the wave phase velocity c such that $c = -\langle u_b \rangle$ (restriction of u to the bottom), the wave travels with speed c in the frame of reference where the mean horizontal velocity is zero at the bed, and where $c > 0$ if the wave travels towards the increasing x direction. For smooth waves, the inequality $u/c < 0$ is true throughout the fluid and at the free surface. A point at which $u = v = 0$ is called a stagnation point.

Consider a steady irrotational wave motion caused by a permanent surface gravity wave. The wave is $2\pi/k$ periodic in the horizontal direction. We suppose that $y = -d$ and $y = \eta(x)$ along with $y = 0$ correspond respectively to the impermeable bottom, the impermeable free surface and the mean water level as appears in Figure 1.1.

In the context of gravity waves, which are waves present in the interface between two media for which the force of gravity and the buoyancy (Archimedes principle) tend to restore equilibrium, the assumption of constant density of the fluid (homogeneity) is physically realistic. Homogeneity implies the equation of mass conservation, which can be formulated by:

$$u_x + v_y = 0 \tag{1.3}$$

We also use the approximation of an inviscid fluid, because the study occurs only for short timescales and lengthscales compared with a wave period or wavelength. Under such hypotheses, we have the following Euler equations:

$$u_t + uu_x + vu_y = -p_x \tag{1.4}$$

$$v_t + uv_x + vv_y = -p_y - g \tag{1.5}$$

where $p(t, x, y)$ is the pressure and g is the gravitational constant of acceleration.

We neglect the surface tension for wavelengths greater than a few centimeters; the typical wavelength of wind-generated waves at the sea is of order 150 meters, and we can suppose that the gravity is the only restoring force exerted on the free surface. The free surface decouples the motion of the water from that of the air [17] using the following dynamic boundary condition:

$$p = 0 \quad \text{on} \quad y = \eta(t, x) \tag{1.6}$$

in which we have normalized the pressure at the surface P with respect to the water density ρ:

$$p = \frac{P - P_{atm}}{\rho} \tag{1.7}$$

We suppose that the same particles are forming the free surface, which yields the kinematic boundary condition

$$v = \eta_t + u\eta_x \quad \text{on} \quad y = \eta(t, x) \tag{1.8}$$

On the bed we have the following condition:

$$v = 0 \quad \text{for} \quad y = -d \tag{1.9}$$

which expresses the fact that the water cannot penetrate the rigid flat bed.

To these equations, we also add the hypothesis of irrotational flows. The vorticity of a two-dimensional flow $\omega = v_x - u_y$ measures the local rotation of a fluid element. In this study, we shall add the following assumption of zero rotational wave motion $\omega = 0$

$\Longleftrightarrow$

$$u_y = v_x \tag{1.10}$$

Denote by ϕ the velocity potential and ψ the stream function, such that

$$u = \phi_x = \psi_y \text{ and } v = \phi_y = -\psi_x \tag{1.11}$$

Consider the complex potential $f = \phi + i\psi$, the complex velocity $w = u - iv$, two holomorphic functions of the variable $z = x + iy$, with $f = f(z)$ and $w = df/dz$. If f and w are holomorphic throughout the fluid domain, the mass conservation and the irrotational character of the flow apply. Since the flow is supposed steady, and the bottom and the surface boundaries,

impermeable, the stream function $\psi = Im(f)$ is constant at $y = -d$, and at $y = \eta$.

The momentum equation is expressed through the Bernoulli condition, which is:

$$2p + 2gy + u^2 + v^2 = B,\ x \in \mathbb{R},\ -d \leq y \leq \eta \tag{1.12}$$

for some Bernoulli constant B, with p denoting the normalized pressure normalized. Combining (1.3), (1.7), (1.8) and (1.13), we obtain:

$$B = \langle u_s^2 + v_s^2 \rangle = \langle u_b \rangle^2 \tag{1.13}$$

where u_b is the restriction of u on the flat bed.

The equation (1.8) as well as the equation (1.3) in [1] give:

$$\langle p \rangle = gd \tag{1.14}$$

where $p(x) = p(x, -d) = p_b(x)$ is the normalized pressure at the bottom.

Chapter 2

Analysis and harmonic analysis results

In this Chapter, we shall recall some classical analysis results which are used in later chapters regarding the reconstruction procedures: since the first procedure relies on fixed point methods we will state the Banach fixed point theorem as well as the Picard-Lipschitz theorem. We shall then underline the main results used in the proofs of Chapter 3's inequalities: the maximum principles are necessary to obtain inequalities related to the pressure p and the velocities u and v.

2.1 Harmonic analysis definitions

Definition 2.1.1 (*Harmonic function*)

A function $f : U \mapsto \mathbb{C}$ is said to be harmonic if it is twice continuously differentiable and satisfies the Laplace condition $\Delta f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$

Definition 2.1.2 (*Subharmonic function*)

Let U be an open subset of $\mathbb{C}$. A function $f : U \mapsto (-\infty, +\infty)$ is said to be subharmonic if f is twice continuously differentiable, and satisfies $\Delta f \leq 0$

Definition 2.1.3 (*Superharmonic function*)

f is said to be superharmonic if $-f$ is subharmonic

2.2 Classical results in analysis

We first start by recalling some results of convergence used later in the section.

2.2.1 Banach fixed point theorem

Theorem 2.2.1 (*Banach fixed point theorem*)

Let (X, d) be a complete non empty metric space. A map $F : X \to X$ is called a contraction if there exists $k \in [0, 1)$ such that $\forall y, y' \in X, d(F(y), F(y')) \leq kd(y, y')$. Then, F admits a unique fixed point $\eta_0 \in X$. This fixed point η_0 can be obtained as the limit of the sequence (y_n) defined as: $y_n = F(y_{n-1})$ starting at an arbitrary point $y_0 \in X$.

2.2.2 Picard-Lipschitz theorem for integral equations

Theorem 2.2.2 (*Picard-Lipschitz theorem*)

Under classical assumptions of lipschitz continuity with respect to the second variable, we have, $\forall n \in \mathbb{N}$, and for some interval J, that the following sequence is well defined: $\eta_{n+1}(x) = \eta_0 + \int_0^x f(x', \eta_n(x'))dx'$

This sequence converges uniformly to the solution of the integral equation.

2.3 Maximum principles for PDE's

2.3.1 Strong maximum principle

Theorem 2.3.1 (*Strong maximum principle*)

Let f be a harmonic function such that $u \in C^2(U) \cap C(\bar{U})$. Then:
(i) $max_U f = max_{\bar{U}} f$
(ii) Furthermore, if U is connected and there exists a point $x_0 \in U$ such that $f(x_0) = max_U f$, then f is constant within U

2.3.2 Hopf's maximum principle for PDEs

The following principle is used to prove some properties of smooth stokes waves in the following section.

Theorem 2.3.2 (*Hopf's maximum principle*)

Let u be a C^2 function on some open set Ω, and assume it satisfies the following inequality: $\sum_{ij} a_{ij}(x)\frac{\partial^2 u}{\partial(x_i)\partial(x_j)} + \sum_i b_i \frac{\partial u}{\partial x_i} \geq 0$, where a_{ij} is a positive definite symmetric matrix and $b_i(x)$ are (locally) bounded coefficients. If u reaches its maximum value M for some $x \in \Omega$, then $\forall x \in \Omega, u(x) = M$

Definition 2.3.3 (*Hodograph transformation*)

The hodograph transform is a technique used to transform certain quasilinear systems of PDE's by reversing the roles of the dependent and independent variables.

Chapter 3

Stokes waves: properties and useful inequalities

3.1 Definition and problem formulation

Definition 3.1.1 (*Stokes wave*)

A Stokes wave is an irrotational smooth solution to the governing equations for which there exists a period $L > 0$ and wave speed c for which η, u, v and P have period L in the x-variable, η depends only on $(x - ct)$, while u, v and P depend only on $(x - ct)$ and y. We also suppose that there exists a single crest and trough per period, and that the profile is decreasing from crest to trough. η, u, and P are symmetric while v is antisymmetric about the crest line.

We take $c > 0$ without loss of generality. We eliminate the time by introducing the change of coordinate system:

$$x' = x - ct \qquad y' = y \tag{3.1}$$

In the moving frame, Bernoulli's law is still true and states that throughout the fluid domain $\Omega = \{(x, y) \in \mathbb{R}^2 : -d < y < \eta(x)\}$, the following expression is a constant:

$$\frac{(u - c)^2 + v^2}{2} + g(y + d) + P \tag{3.2}$$

.

Using (1.4) and (1.5), we define the stream function $\Psi(x, y)$ (up to a constant) by:

$$\Psi_y = u - c, \quad \Psi_x = -v \tag{3.3}$$

Ψ is harmonic in Ω by (1.4) whereas (1.8) and (1.9) guarantee that Ψ is constant on both boundaries of Ω. For instance, we can suppose $\Psi = 0$ on $y = \eta(x)$ and also that $\Psi = m$ on $y = -d$.

Therefore, $\Psi(x, y)$ is of the form:

$$\Psi(x, y) = m + \int_{-d}^{y} (u(x, s) - c)ds \tag{3.4}$$

Ψ has a period of L in the x-variable. We could therefore express the boundary condition (1.7) in an equivalent manner, for a Stokes wave, as a smooth solution $(\eta(x), \Psi(x, y))$, periodic and even function in the $x-$variable such that η has one maximum and one minimum per period. It can be viewed as a solution of the free boundary value problem:

$$\begin{cases} \Delta\Psi = 0 & \text{in} \quad -d < y < \eta(x) \\ \frac{|\Psi|^2}{2} + g(y + d) = Q & \text{on} \quad y = \eta(x) \\ \Psi = 0 & \text{on} \quad y = \eta(x) \\ \Psi = m & \text{on} \quad y = -d \end{cases} \tag{3.5}$$

In these equations, m and Q are physical constants . We also have $Q > 0$ since $\eta(x) + d > 0$. It also follows from [29,32] that $m \neq 0$ (unless $\Psi \equiv 0$, in which case the surface is flat. Suppose $m > 0$, so that the strong maximum principle [14] applies to Ψ and to Ψ_y, which gives:

$$\Psi_y = u - c < 0 \quad \text{in} \quad \overline{\Omega} \tag{3.6}$$

(unless $\Psi \equiv 0$).

According to experimental data, the horizontal motion of individual water particles is slower than the propagation speed of the wave. In the moving frame, the wave speed $c > 0$ itself does not appear in the previous system of equations.

At the bottom, we shall define the mean current , by:

$$\kappa = \frac{1}{L} \int_0^L u(x, -d)dx < c \tag{3.7}$$

where the inequality follows from (2.10) Since u is harmonic, it follow that also equals the average of u on any horizontal line below the trough. We shall apply Green's theorem to observe it. For all y_0 below the wave trough, we have:

$$0 = \int_{-d}^{y_0} \int_0^L (u_y - v_x)dxdy = \int_0^L u(x, y_0)dx - \int_0^L u(x, -d)dx \tag{3.8}$$

Therefore,

$$0 = \int_{-d}^{y_0} \int_0^L (u_y - v_x) dx dy = \int_0^L u(x, y_0) dx - \int_0^L u(x, -d) dx \tag{3.9}$$

3.2 Properties of Stokes waves

We shall now state and prove some results about the pressure function p and the horizontal velocity u. To this aim, introduce the following notations:

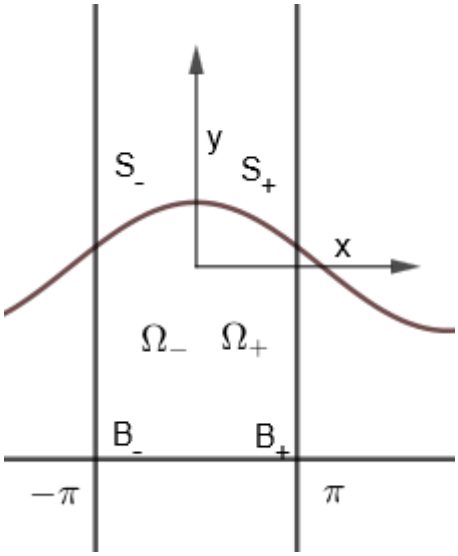


Figure 3.1: Dividing the plane into sets

Consider the following sets:
$\Omega_- = \{(x, y) \in \mathbb{R}^{\nvDash}, \quad x \in (-\pi, 0), \quad -d < y < \eta(x)\}$
$\Omega_+ = \{(x, y) \in \mathbb{R}^{\nvDash}, \quad x \in (0, \pi), \quad -d < y < \eta(x)\}$

The two halves of the free surface S in one period of the fluid domain Ω:
$S_+ = \{(x, y) \in \mathbb{R}^{\nvDash}, \quad x \in (0, \pi), \quad y = \eta(x)\}$
$S_- = \{(x, y) \in \mathbb{R}^{\nvDash}, \quad x \in (-\pi, 0), \quad y = \eta(x)\}$

Introduce also the two halves of the flat bed:
$B_+ = \{(x, y) \in \mathbb{R}^{\nvDash}, \quad x \in (0, \pi), \quad y = -d\}$
$B_- = \{(x, y) \in \mathbb{R}^{\nvDash}, \quad x \in (-\pi, 0), \quad y = -d\}$

Before stating any results, we would like to present here the technique of the hodograph conformal transformation, which will allow us to reformulate the system (3.5) differently. We set:

$$\boxed{\begin{aligned} q &= -\Psi(x,y) \\ p &= -\Phi(x,y) \end{aligned}} \tag{3.10}$$

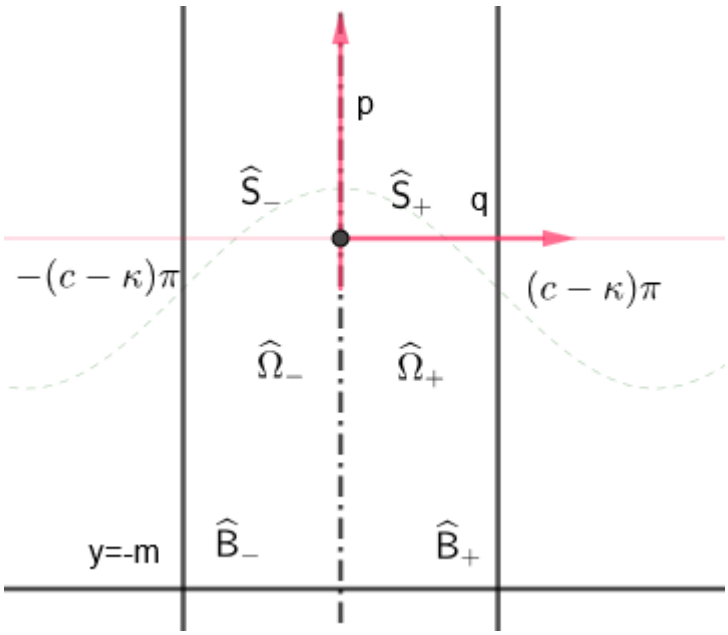


Figure 3.2: Effects of the change of variables on the plane regions

This transformation transforms the free boundary problem (3.5) into a non linear boundary problem for the harmonic function $h(q,p) = y + d$ in a fixed rectangular domain. The images of the previously introduced sets Ω_+, S_+ and B_+ are transformed respectively to:

$$\begin{aligned} \hat{\Omega}_+ &= (0,(c-\kappa)\pi) \times (-m,0) \\ \hat{S}_+ &= (0,(c-\kappa)\pi) \times \{0\} \\ \hat{B}_+ &= (0,(c-\kappa)\pi) \times \{-m\}. \end{aligned} \tag{3.11}$$

(And similarly for Ω_-, S_- and B_-).

The system (3.5) becomes:

$$\begin{cases} \Delta_{q,p} h = 0 & \text{for} \quad -(c-\kappa)\pi < q < (c-\kappa)\pi, -m < p < 0 \\ h = 0 & \text{on} \quad p = -m \\ 2(Q - gh)(h_q^2 + h_p^2) = 1 & \text{on} \quad p = 0 \end{cases} \tag{3.12}$$

where the function h is even and $(c-\kappa)2\pi-$periodic in the q variable.

By classical chain rules, we deduce the following relations:

$$\begin{aligned} \partial_q &= h_p \partial_x + h_q \partial_y \\ \partial_p &= -h_q \partial_x + h_p \partial_y \end{aligned} \tag{3.13}$$

We also notice the following:

$$\begin{aligned} \partial_x &= (c-u)\partial_q + v\partial_p \\ \partial_y &= -v\partial_q + (c-u)\partial_p \end{aligned} \tag{3.14}$$

Furthermore, we have:

$$\begin{aligned} h_q &= -\frac{v}{(c-u)^2 + v^2} = -\frac{\partial x}{\partial p} = \frac{\partial y}{\partial q} \\ h_p &= \frac{c-u}{(c-u)^2 + v^2} = \frac{\partial x}{\partial q} = \frac{\partial y}{\partial p} \end{aligned} \tag{3.15}$$

3.2.1 Minimizing the pressure function p

Let us now state the two main results of this section, regarding the variations of the pressure (as a function of x and y), but also results regarding the horizontal velocity u and the vertical velocity v.

Theorem 3.2.1 (*Pressure variations and inequality*)

(i) $\forall y \in [-d, \eta_0], \quad p_y(0, y) < 0$.
The pressure p attains its minimum value only at the free surface, where it is zero.
(ii) In the moving frame, we have $p_x < 0$ in $\overline{\Omega}_+$ and $p_x > 0$ except on the crest and trough lines where $p_x = 0$.

Theorem 3.2.2 (*Horizontal velocity inequality*)

(i) Along every streamline $\{\Psi = k\}$ with $\quad 0 \leq k \leq m$ the horizontal velocity u is strictly increasing in $\overline{\Omega}_+$ and strictly increasing in $\overline{\Omega}_-$ as a function of x.

(ii) $\forall y \in [-d, \eta_0] \quad u_y(0, y) > 0$
$\forall x, \quad \partial_x u_x(0, \eta_x) < 0$

Theorem 3.2.3 (***Vertical velocity inequality***)

$\forall y \in [-d, \eta_0], \quad \forall x \in (0, L/2), \quad v(x, y) > 0$

Before proving any theorems, we would like to apply Hopf's maximum principle in order to obtain the following facts regarding the functions u and v.

General facts about u and v

$$\begin{aligned} &\forall x \in (0, \pi) \\ &v(x, \eta(x)) > 0 \quad \text{and} \quad \partial_x[u(x, \eta(x))] < 0 \end{aligned} \tag{3.16}$$

We also have the following inequalities which account for the variations of u:

$$\begin{aligned} &u_y(0, y) > 0 \quad \text{for} \quad y \in (-d, \eta(0)) \\ &u_x(x, -d) < 0 \quad \text{for} \quad x \in (0, \pi) \\ &u_y(\pi, y) < 0 \quad \text{for} \quad y \in (-d, \eta(\pi)) \end{aligned} \tag{3.17}$$

First start by normalizing the wavelength $L = 2\pi$, and assume that in the moving frame, the wave crest starts at the point $(0, \eta_0)$ and the wave troughs at the points of the form $(\pm\pi, \eta(\pm\pi))$.
The crest line is $\{x = 0, \quad -d < y < \eta(0)\}$, while the trough lines are $\{x = \pm\pi, \quad -d < y < \eta(\pm\pi)\}$. The wave profile is symmetric about the crest and $\eta_x \leq 0$ on S_+.

From (2.4), we have that $v = (u - c)\eta_x \quad \text{for} \quad y = \eta(x)$ so that (2.10) holds and the assumption $\eta_x(x) \leq 0$ for $x \in [0, \pi]$, which ensures $v(x, \eta(x)) \geq 0$ for $x \in [0, \pi]$, with equality only when $\eta_x(x) = 0$. For $x \in [0, \pi], \quad v(x, -d) = 0$ by (2.6). Furthermore, $v = 0$ on the vertical segments $x = 0$ and $x = \pi$, since v is odd and periodic in the x-variable throughout Ω.

If the free surface is not flat, there is at least a point on S_+ where $v > 0$. Since $v \geq 0$ on the boundary of Ω_+, the maximum principle yields:

$$v > 0 \quad \text{in} \quad \Omega_+ \quad \text{while} \quad v < 0 \quad \text{in} \quad \Omega_- \tag{3.18}$$

which reminds of the fact that v is odd in the x-variable. On the other hand, by (2.2) we have that P is superharmonic in Ω, since

$$\Delta P = -\Psi_{xx}^2 - 2\Psi_{x,y}^2 - \Psi_{yy}^2 \leq 0 \tag{3.19}$$

.

Since $P_y = -g < 0$ on $y = -d$ by (2.2) and (2.5), we can apply Hopf's maximum principle which implies that the minimum of P is attained only at the free surface $\{y = \eta(x)\}$, where $P = 0$ in view of (2.3).

By Hopf's principle, $P_x(x, \eta(x)) \leq 0$ along S_+, with $P_x(x, \eta(x)) < 0$ when $\eta_x(x) < 0$ for $x \in (0, \pi)$. But $\eta_x(x)$ is necessarily strictly negative for $x \in (0, \pi)$ even if it can be allowed to be 0 at some points theoretically.

If for some $x_0 \in (0, \pi)$ we have $\eta_x(x_0) = 0$, then $v(x_0, eta(x_0)) = 0$ and $v_y(x_0, \eta(x_0)) < 0$. By (2.1) and (2.2), we find that $P_x = -(c - u)v_y > 0$ at $(x_0, \eta_(x_0))$ and the opposite inequality is also true.

Therefore, we conclude that $\eta_x < 0$, $v > 0$ and $P_x < 0$ at all points on S_+. Since $(c - u)_x(u(x, \eta(x))) = P_x(x, \eta(x))$ by (2.2) and (2.4) we can write:

$$v(x, \eta(x)) > 0 \quad \text{and} \quad \partial_x[u(x, \eta(x))] < 0 \forall x \in (0, \pi) \tag{3.20}$$

Furthermore, u is strictly decreasing as we go from crest to trough along the lower three sides of Ω_+, which implies the following strict inequalities:

$$\begin{aligned} &u_y(0, y) > 0 \text{ for } y \in (-d, \eta(0)) \\ &u_x(x, -d) < 0 \text{ for } x \in (0, \pi) \\ &u_y(\pi, y) < 0 \text{ for } y \in (-d, \eta(\pi) \end{aligned} \tag{3.21}$$

along these three lines.

Indeed we have $v = 0$ along the three sides so that (3.3) follows from Hopf's principle by using (2.1), (2.6) and (3.1). By irrotationality, we can define the velocity potential $\Phi(x, y)$ by $\Phi_x = u - c$, $\Phi_y = v$, with $\Phi = 0$ on the crest line. That is

$$\Psi(x, y) = \int_0^x [u(l, -d) - c]dl + \int_- d^y v(x, s)ds \tag{3.22}$$

.

Therefore, $\Psi(x, y) + (c - \kappa)x$ has period of 2π in x. Moreover, Phi is odd in the x variable and vanishes at $x = 0$, while:

$\Phi(2, y) = \int_0^2 [u(l, y) - c]dl = -(c - \kappa)2\pi n$ for any y below the trough level and for any integer n by (2.12).

We shall now prove the theorem.
We can first derive some useful formulas. If $(x, y(x))$ is the parametric equation of some streamline, then $\partial_x[\Psi(x, y(x))] = 0$, by definition so that:

$$\frac{dy}{dx} = y' = \frac{-\Psi_x}{\Psi_y} = \frac{v}{u-c} = \frac{h_q}{h_p} \tag{3.23}$$

.

By (3.6), it follows that:

$$\partial_x[u(x, y(x))] = u_x + y'u_y = u_x + \frac{h_q}{h_p}u_y = \frac{1}{h_p}u_q \tag{3.24}$$

On the other hand, by (2.2) and (3.8), we get

$$P_x = (c-u)u_x - vu_y = \frac{h_p u_x + h_q u_y}{h_q^2 + h_p^2} = \frac{1}{h_q^2 + h_p^2}u_q \tag{3.25}$$

Since u and v are harmonic in (x, y)variables, they remain harmonic in the (q, p) variables since (3.4) is a conformal change of variables. So u_q is also harmonic. Alternatively, u_q is harmonic because $u_q = Re(-\zeta''(z)/\zeta'(z))$ where $\zeta = \Phi + i\Psi$ and $z = x + iy$.

Let us consider the restriction of u_q to the rectangle $\hat{\Omega_+}$. By (3.6) and (3.8) we have:

$u_q = \frac{(c-u)u_x - vu_y}{(c-u)^2+v^2}$. Along the lateral sides of $\hat{\Omega}_+$ which are the images under (3.4) of the crest and trough lines, we have:

$$u_q = \frac{u_x}{(c-u)} = \frac{-v_y}{c-u} = 0 \quad \text{for} \quad q = 0 \quad \text{and} \quad q = (c-\kappa)\pi \tag{3.26}$$

since $v = v_y = 0$ on these sides. On the lower boundary $\hat{B_+} = \{p = -m\}$ of the rectangle $\hat{\Omega_+}$, the image of the flat bed, we also know by (2.5) that v=0, so that:

$$u_q = \frac{1}{c-u}u_x < 0 \text{if} p = -m \quad \text{and} \quad q \in (0, (c-\kappa)\pi) \tag{3.27}$$

by (3.3). On the top boundary $\hat{S}_+ = \{p = 0\}$, the image of the free surface from crest to trough, we have:

$$\begin{aligned} u_q &= \frac{[(c-u)u_x - vu_y}{(c-u)^2 + v^2} \\ &= \frac{c-u}{(c-u)^2+v^2}\partial_x[u(x, \eta(x))] < 0 \end{aligned} \tag{3.28}$$

for $p = 0, q \in [0, (c-\kappa)\pi]$, by (2.4), (2.10) and (3.2). Therefore, by (3.11) - (3.13) and the strong maximum principle applied to u_q, we deduce that $u_q < 0$ in the open rectangle $\hat{\Omega}_+$.

But then, (3.9) and $h_p > 0$ ensure that u is strictly decreasing in x along any streamline in $bar\Omega_+$. Since u is even in the $x - variable$, u is strictly increasing in x along any streamline in $O\bar{m}ega_-$

We shall now prove the theorem (pressure).

Part (i) of the theorem follows from (3.10). For part (ii), along a trough line we have $v = 0$. From (2.2) and (2.6) we obtain:

$$P_y + g = (c-u)v_x = (c-u)u_y < 0 \tag{3.29}$$

along $\{x = \pi\}$, because of (2.10) and (3.3).

We claim that:

$$\begin{aligned} &\partial_x(\{c - u(x, \eta(x))\}v(x, \eta(x))) \\ &= -u_x v - u_y v \eta_x + (c-u)v_x + (c-u)v_y\eta_x = v_x(1+\eta_x^2)(c-u). \end{aligned} \tag{3.30}$$

By differentiating the Bernoulli boundary condition in (2.9), we obtain:

$$\begin{aligned} 0 &= (u-c)(u_x + u_y\eta_x) + v(v_x + v_y\eta_x) + g\eta_x \\ &= (u-c)u_x + u_y v + vv_x + (u-c)\eta_x^2 v_y + g\eta_x \end{aligned} \tag{3.31}$$

by (2.4).

So (2.1) and (2.6) yield:

$$v_y(c-u)(1-\eta_x^2) + 2vv_x + g\eta_x = 0 \tag{3.32}$$

along S.

First consider $x \in (0, \pi)$, Dividing (3.17) by $\eta_x < 0$, we obtain:

$$(c-u)[\frac{v_y}{\eta_x}(1-\eta_x^2) - 2v_x] + g = 0 \tag{3.33}$$

by using (2.4). From the second inequality in (3.2) and (2.1) and (2.6), we have:

$$\frac{v_y}{\eta_x} < v_x \tag{3.34}$$

along S_+

Therefore, (3.16) and (3.19) imply (3.15) on S_+ with a strict inequality. Inequality (3.15) follows because its left side is even.

We shall now prove that $P_y < 0$ everywhere. Since (3.15) is true, we consuder the function $f(x,y) = (c-u)v - gx$ which is harmonic in Ω since

$$\Delta f = (c-u) - v\Delta u - 2u_x v_x - 2u_y v_y = 0 \tag{3.35}$$

due to (2.1) and (2.6).
Consider the function f of the variables $(q,p) \in \bar{R} = [-(c-\kappa)\pi, (c-\kappa)\pi] \times [-m, 0]$. We shall apply the maximum principle to f_q in the open rectangle R. Bu (2.1), (2.6), (3.6) and (3.8), compute:

$$f_q = v_x - \frac{g(c-u)}{(c-u)^2 + v^2} \tag{3.36}$$

because two terms cancel. Since $v = v_x = 0$ on the bottom $\hat{B} = \{p = -m\}$, we obtain:

$$f_q = \frac{-g}{c-u} < 0 \tag{3.37}$$

on $\hat{B}$, again using the fact (2.10) that $u < c$. For the lateral sides of R, we have by (3.20), (2.4), (3.15), (3.16) and (2.10) that:

$$f_q = \frac{\partial_x\{f(x,\eta(x))\}}{(c-u)(1+\eta_x^2)} \leq 0 \tag{3.38}$$

Since f is harmonic in the $(x,y) - variables$, it is also harmonic in the (q,p) variables and so does f_q. By the maximum principle applied to f_q, as well as the inequalities (3.21)$-$(3.23), we deduce $f_q < 0$ in the open rectangle R. By (3.20), we obtain the inequality:

$$v_x < \frac{g(c-u)}{(c-u)^2 + v^2} \quad \text{in} \quad R \tag{3.39}$$

On the other hand, we had already established that:

$$0 > u_q = \frac{(c-u)u_x - vu_y}{(c-u)^2 + v^2} \quad \text{in} \quad \Omega_+ \tag{3.40}$$

By (2.1), (2.6), (2.10) this inequality is equivalent to:

$$v_y > \frac{v}{u-c} v_x \quad \text{in} \quad \Omega_+ \tag{3.41}$$

By (2.2), we infer that:

$$P_y = -g + (c-u)v_x - vv_y < -g + \frac{v_x}{c-u}[(c-u)^2 + v^2] < 0 in \Omega_+ \tag{3.42}$$

where the first equality is from (3.25) and the second inequality is from (3.24) and the second inequality is from (2.24) recalling (3.1). This proves the desired inequality throughout the open set Ω_+. As for the boundary of Ω_+, we proved in (3.14) that $P_y < 0$ on the trough lines. On teh crest line, $v = 0$ so that by (2.2) and (3.20) we have:

$$P_y = (c - u)v_x - g = (c - u)f_q < 0 \tag{3.43}$$

since $f_q < 0$ in R, and (2.10) holds. On the flat bed B_+, (2.2) and (2.5) ensure that $P_y = -g < 0$. Finally on the free surface S_+, the superharmonic function P attains its minimum $P = 0$, so by Hopf's maximum principle, $P_y < 0$ there as well. By evenness, $P_y < 0$ throughout the fluid.

Chapter 4

Classical procedure for the recovery of the wave profile

We shall now detail a method to obtain η as the solution to the equation (3.12). The proof lies in three key steps detailed separately:

- **Step 1** Determination of the wave amplitude η_0
- **Step 2** Finding a suitable approximation b of the Bernoulli constant B
- **Step 3** Computing the surface elevation η

4.1 Determination of wave amplitude η_0

The first step is to use known p to determine $\mathcal{B}$, for instance by using (3.10). Let us define $\forall y \in [-d, \eta_0]$, the function F for which η_0 is a fixed point (we shall check that it exists below): $F(y) = (Re(\mathcal{B})(0, y)/g - d$.

Working at the crest line $x = 0$, where $v = 0$ we have $\mathcal{B} = p + g(y + d)$. F becomes:

$$F(y) = y + p(0, y)/g = (B - u^2(0, y))/2g \tag{4.1}$$

.

Let us now check that F is indeed a contraction.

$$\begin{aligned} &\forall y, y' \in [-d, \eta_0], \\ &|F(y) - F(y')| = |y - y' + p(0, y)/g - p(0, y')/g| \\ &= \frac{1}{2g}|u(0, y') - u(0, y)| \\ &\leq \frac{1}{2g}|y' - y| \end{aligned} \tag{4.2}$$

(where we have used theorem 2.2.3). Therefore, F is indeed a contraction.

By Banach fixed point theorem, F admits a unique fix point η_0, obtained as a limit of the sequence $y_n = F(y_{n-1})$ with any initial condition $y_0 \in [-d, 0]$.

By theorem 2.2.1, p attains its minimum of 0 only at the free surface, we get that the sequence (y_n) is clearly increasing. This proof provides a scheme to determine η_0 by an iterative procedure.

4.2 Finding an approximation b of the Bernoulli constant B

The second step is to use p to determine b, which is an approximation of the Bernoulli constant B. By (2.11) and (3.1), $\forall y \in [-d, \eta_x]$,

$$B - gy - (Re(\mathcal{B}))(x, y) + gd = p(x, y) + u^2(x, y) \geq u^2(x, y) \geq u(0, \eta_0) \quad (4.3)$$

(where we have used theorem 2.2.3 and the fact that there are no stagnation points, ie that $u(x, y) < 0$). Since $u^2(0, \eta_0) = B - g\eta_0 - (Re(\mathcal{B})(0, \eta_0) + gd > 0$, we conclude that:

$$B - gy - (Re(\mathcal{B}))(x, y) + gd > B - g\eta_0 - (Re(\mathcal{B})(0, \eta_0) + gd > 0 \quad (4.4)$$

For b such that:

$$b > g(\eta_0 - d) + (Re(\mathcal{B})(0, \eta_0) \quad (4.5)$$

we have:

$$\begin{aligned} & b - gy - (Re(\mathcal{B})(x, y) + gd \\ & = (b - B) + B - gy - (Re(\mathcal{B})(x, y) + gd \\ & \geq (b - B) + B - g\eta_0 - (Re\mathcal{B}(0, \eta_0)) + gd \\ & \geq b - g(\eta_0 - d) + Re(\mathcal{B})(0, \eta_0) > 0 \end{aligned} \quad (4.6)$$

Therefore, if b is such that (4.5) is satisfied, and such that

$$b \approx g(\eta_0 - d) + Re(\mathcal{B})(0, \eta_0) \quad (4.7)$$

we obtain that $b < B$.

Finally, we have obtained an estimation of B using p.

4.3 Computing the surface elevation η

For the constant b we have previously computed, we shall apply the iterative procedure defined below. $\forall n \geq 1$, $\forall x \in [0, L/2]$:

$$\eta^b_{n+1}(x) = \eta_0 + \int_0^x f(x', \eta^b_n(x'))dx' \tag{4.8}$$

where $f_n(x', \eta^b_x) = \frac{Im(\mathcal{B}(x', \eta^b_n(x')}{b - g\eta^b_n(x') - (Re(\mathcal{B}(x', \eta^b_n(x')) + gd}$.

Remark: The b exponent implies the dependency on the b computed in the previous section.

We therefore have to check that the above integrand is smooth, which is the case as long as the denominator is non 0. It remains to check that f is lipschitz continuous. Applying theorem 2.1.2, we obtain that this sequence converges uniformly on $[0, L/2]$ to a certain function, $\eta^b(x)$.

Using theorem 2.2.4, in the equation (3.12) used with (2.8) and (3.13) we notice that for a given couple (x, η), the RHS is a strictly increasing function of B. This shows that:

$$b_1 < b_2 \leq B \implies \eta^{b_1}(x) < \eta^{b_2}(x) \leq \eta^B(x) \tag{4.9}$$

There exists therefore a unique value of b for which we have $< \eta^b >= 0$. It is sufficient to sample several values of b with the previous constraint, until we obtain a η_x with zero mean. Let us reformulate this problem using the average operator.

$$\eta^B(x) = \eta_0 + \int_0^x \frac{Im(\mathcal{B}(x', \eta^b_n(x')dx'}{b - g\eta^B(x') - (Re(\mathcal{B}(x', \eta^b(x')) + gd} \tag{4.10}$$

.

Let us take the mean on $[0, L/2]$ for a b such that $< \eta^B >= 0$, and integrate by parts. We obtain:

$$\begin{aligned} \langle \eta^B(x) \rangle &= \langle \eta_0 \rangle + \langle \int_0^x \frac{Im(\mathcal{B}(x', \eta^b_n(x')dx'}{b - g\eta^B(x') - (Re(\mathcal{B}(x', \eta^B(x')) + gd} \rangle \\ \eta_0 &= -\frac{2}{L} \int_0^{L/2} \int_0^x \frac{Im(\mathcal{B}(x', \eta^b_n(x')}{b - g\eta^B(x') - (Re(\mathcal{B}(x', \eta^B(x')) + gd} dx dx' \end{aligned} \tag{4.11}$$

Integrating by parts yields immediately:

$$\eta_0 = -\frac{2}{L}\int_0^{L/2}(\frac{L}{2}\int_0^{L/2}\frac{Im(\mathcal{B}(x',\eta_n^b(x')}{b-g\eta^B(x')-(Re(\mathcal{B}(x',\eta^B(x'))+gd}dx'+$$
$$\int_0^{L/2}x\frac{Im(\mathcal{B}(x',\eta_n^b(x')}{b-g\eta^B(x')-(Re(\mathcal{B}(x',\eta^B(x'))+gd}dx' \tag{4.12}$$

$$\boxed{\eta_0 = \frac{2}{L}\int_0^{L/2}(\frac{-L}{2}+x)\frac{Im(\mathcal{B}(x',\eta_n^b(x')}{b-g\eta^B(x')-(Re(\mathcal{B}(x',\eta^B(x'))+gd}dx'}$$

Chapter 5

A new reconstruction procedure

In this section, we shall present another way to recover the surface as described by D. Clamond in 'New exact relations for easy recovery of steady wave profiles from bottom pressure measurements'. This method consists of two main steps:

- **Step 1** Introduction of some holomorphic functions $\mathcal{B}$ and then $\mathcal{D}$ and extracting the wave profile η as a solution to an implicit equation, which involves $\mathcal{D}$.
- **Step 2** Finding a suitable approximation B of the Bernoulli (two methods). We will emphasize the method using the holomorphic functions (which has been implemented numerically in the following chapter).

5.1 Implicit equation involving η

Let us define the following function $\mathcal{B}$:

$$\begin{aligned}\mathcal{B}(z) &= \frac{1}{2}B + gd - \frac{1}{2}w^2(z) \\ &= \frac{1}{2}B + gd - \frac{1}{2}(u^2 - v^2) + iuv\end{aligned} \tag{5.1}$$

Notice that this function has zero imaginary part at the flat bed $y = -d$ with real part equal to $p(x) = \mathcal{B}(x - id) = gd + (B - u_b^2)/2$. This means that $\mathcal{B}(z) = p(z + id)$. p coincides with $Re(\mathcal{B})$ only at $y = -d$ (because p is not harmonic).

Using (1.13) and the fact that the surface is impermeable, at the free surface $y = \eta(x)$, $v_s = u_s\eta_x$, we have therefore:

$$
\begin{aligned}
&(u_s - iv_s)^2 = (1 - i\eta_x)^2 u_s^2 = (1 + \eta_x^2) u_s^2 (1 - i\eta_x)/(1 + i\eta_x) \\
&= (u_s^2 + v_s^2)(1 - i\eta_x)/(1 + i\eta_x) \\
&= (B - 2g\eta)(1 - i\eta_x)/(1 + i\eta_x)
\end{aligned} \tag{5.2}
$$

Multiplying by $(1 + i\eta_x)$ and using the definition of $\mathcal{B}$, we obtain:

$$
\begin{aligned}
&g\eta(1 - i\eta_x) + iB\eta_x \\
&= (\mathcal{B}_s - gd)(1 + i\eta_x)
\end{aligned} \tag{5.3}
$$

where $\mathcal{B}_s = \mathcal{B}(z_s) = \mathcal{B}(x + i\eta(x))$. By taking real and imaginary parts in (5.3), we obtain the two following equations:

$$
g\eta = Re\mathcal{B}_s - gd - \eta_x Im(\mathcal{B}_s) = \frac{1}{2}(B - u_s^2 - v_s^2) \tag{5.4}
$$

$$
(B - g\eta)\eta_x = (Re(\mathcal{B}_s) - gd)\eta_x + Im(\mathcal{B}_s) = \frac{1}{2}(B + u_s^2 + v_s^2)\eta_x \tag{5.5}
$$

in which we have replaced v_s by $\eta_x u_s$ and used the definition of $\mathcal{B}$.

We define now $\mathcal{D}$ another holomorphic function:

$$
\mathcal{D}(z) = \int_{z_0}^{z} (\mathcal{B}(z') - gd)dz' = \frac{1}{2}\int_{z_0}^{z} (B - w(z'))dz' \tag{5.6}
$$

z_0 is a constant that we will first choose at the origin of the free surface for $x = 0$ and $y = a$ which means $z_0 = ia$ (This choice is arbitrary, and could be replaced by any $z_0 = x_0 + i\eta(x_0)$).

We let $z(x') = x' + i\eta(x')$ as x' goes from 0 to x, we apply the complex integration formula: $dz' = (1 + i\eta_x(x'))dx'$:

$$
\mathcal{D}_s(z) = \int_0^x (\mathcal{B}(x + i\eta_{x'}) - gd)(1 + i\eta_x(x')dx' \tag{5.7}
$$

Using (3.3) by replacing $\mathcal{B}_s - gd$, $\mathcal{B}_s$ being the value at the surface of $\mathcal{B}$, we obtain:

$$
\begin{aligned}
\mathcal{D}_s(z) &= \int_0^x (g\eta(1 - i\eta_x) + iB\eta_x dx' \\
\mathcal{D}_s(z) &= \int_0^x (g\eta(x')dx' - ig\int_0^x \eta\eta_x dx' + iB\int_0^x \eta_x dx' \\
\mathcal{D}_s(z) &= \int_0^x (g\eta(x')dx' - ig/2[\eta^2]_0^x + iB[\eta]_0^x \\
\mathcal{D}_s(z) &= \int_0^x (g\eta(x')dx' - ig/2[\eta^2(x) - a^2] + iB[\eta(x) - a]
\end{aligned} \tag{5.8}
$$

We have finally obtained the formula given by:

$$\boxed{\mathcal{D}_s(z) = \int_0^x (g\eta(x')dx' + i[\eta^2(x) - a][B - \frac{1}{2}ga - \frac{1}{2}g\eta(x)]} \tag{5.9}$$

We shall now determine η by taking the imaginary parts of the previous equation, and solving the quadratic equation in η. We obtain two solutions, of which we keep only the one such that $\eta(0) = a$:

$$\boxed{\eta = g^{-1}[B - \sqrt{(B - ga)^2 - 2gIm(\mathcal{D}_s)}]} \tag{5.10}$$

In order to express $Im(\mathcal{D}_s)$, we use (5.8) to obtain a local algebraic equation determining η. We can check that this last equation is exactly equivalent to the one we have derived in the previous chapter. Differentiating with respect to x, and using the previous equivalence we obtain:

$$\boxed{\eta_x = \frac{[Im(\mathcal{D}_s)]_x}{\sqrt{(B - ga)^2 - 2gIm(\mathcal{D}_s)}} = \frac{Im(\mathcal{B}_s + (Re(\mathcal{B}_s) - gd)\eta_x}{B - g\eta}} \tag{5.11}$$

5.2 Determination of the Bernoulli constant

The next step is to determine the Bernoulli constant B, and to this end we have two possible methods.

- **Method 1** Using a non local relation

Using the average operator and (1.1) on the imaginary part of equation (5.11), and using the zero mean property, we get:

$$2aB = g(a^2 - \langle\eta^2\rangle) - 2(Im(\mathcal{D}_s) \tag{5.12}$$

. This relation is non local because it involves integration.

- **Method 2** Using two implicit relations and writing B as a function of the two solutions.

Let us introduce the two following quantities a and b', defined as the possible solutions of the following implicit equations:

$$a = g^{-1}(Re(\mathcal{B}(ia)) - d \tag{5.13}$$

$$b' = d - g^{-1}Re(\mathcal{B}(\pi/k - ib')) \tag{5.14}$$

By using equation (6.4) at the point $x = \pi/k$, $\eta = -b$, we obtain the following expression for B:

$$B = \frac{1}{2}(a - b') - (a + b)^{-1} Im(\mathcal{D}(\pi/k - ib')) \tag{5.15}$$

We can obtain $\mathcal{D}$ by using $\mathcal{B}$ which in turn is determined by measuring p at the bed. We will discuss how to obtain these quantities numerically in the following section.

Chapter 6

Numerical results

In this section, we provide a practical method of recovering the surface elevation using experimental pressure measurements at the bed. It is based on Chapter 5 as it is more convenient to implement numerically.

We suppose we have sample values of the pressure p at certain points of the x-axis that are evenly spaced. We use a Fourier polynomial of order N to approximate this pressure as follows:

$$p(x) = gd + \sum_{|n|>0}^{N} p_n exp(inkx) \tag{6.1}$$

This allows us to determine $\mathcal{B}$, which is then given by:

$$\mathcal{B}(z) = gd + \sum_{|n|>0}^{N} p_n \frac{exp(inkx)}{exp(nkd)} \tag{6.2}$$

This allows us to subsequently derive $\mathcal{D}(z)$, with the formula:

$$\mathcal{D}(z) = \sum_{|n|>0}^{N} \frac{p_n}{ink} \frac{exp(inkz) - exp(-nka)}{exp(nkd)} \tag{6.3}$$

It is therefore possible to compute B obtained from (**??**). Then we have the formula (**??**) which allows us to extract η:

$$\eta = g^{-1}[B - \sqrt{(B - ga)^2 - 2gIm(\mathcal{D}_s)}] \tag{6.4}$$

6.1 Wave recovery graphs

We provide the algorithm along with some illustrations of the rate of converge (depending on the order of the Fourier polynomial used to approximate the three above functions) below. Moreover, let Δ_1 (resp. Δ_2) be the error on the values of a (resp b) between the approximation and the real wave amplitude. Let us observe the results for $N = 5, 10, 20, 40, 100$ and how the errors correlate with the order of the Fourier polynomial.

Figure 6.1: N=5 $\Delta_1 = 2.6 \times 10^{-4}$ $\Delta_2 = 1.4 \times 10^{-5}$

Figure 6.2: N=10 $\Delta_1 = 7.3 \times 10^{-7}$ $\Delta_2 = 1.3 \times 10^{-9}$

Figure 6.3: N=20 $\Delta_1 = 6.0 \times 10^{-5}$ $\Delta_2 = 2.6 \times 10^{-9}$

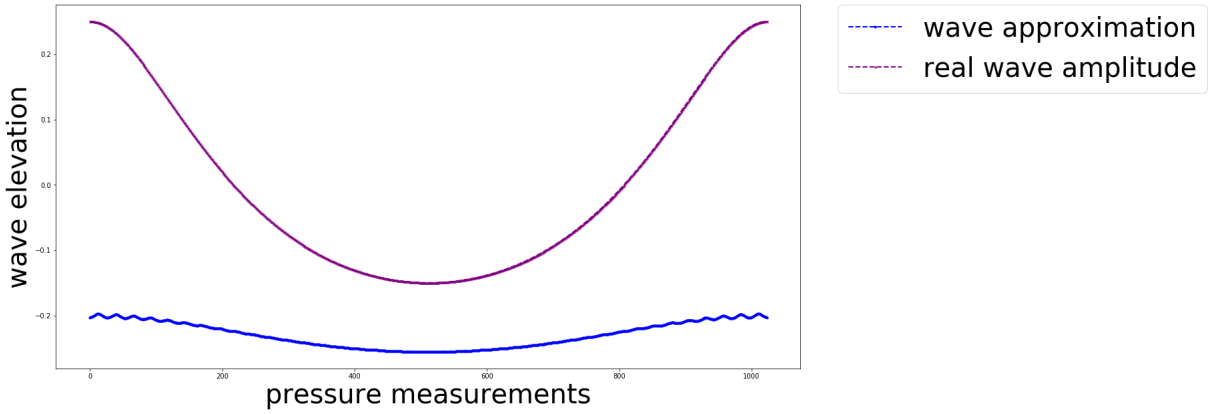


Figure 6.4: N=40 $\Delta_1 = 0.45$ $\Delta_2 = 0.11$

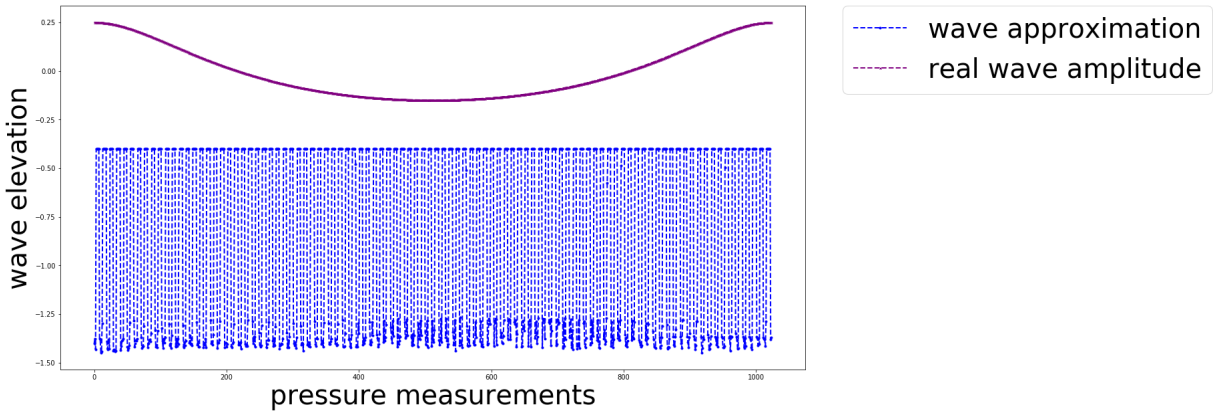


Figure 6.5: N=100 $\Delta_1 = 0.65$ $\Delta_2 = 1.3$

Chapter 7

Conclusion

We have observed two methods to recover the wave profile: one is based on iterative fixed point methods which only grant an approximation of η. The second procedure provides an exact formula which not only can be checked to be a solution to the first problem, but it can also be easily implemented numerically; it is worth mentioning in this regard that the order of the Fourier approximation chosen plays a crucial role in recovering a satisfactory approximation of the surface elevation. The order of the Fourier polynomial should not be too large for convergence to occur.

A possible generalization could be to extend the physical hypotheses presented in Chapter 1. For instance, in this master's thesis we have only dealt with smooth Stokes waves for which the flow does not have any stagnation points (in other words, $u(x, y) < 0$ for any $x \in \mathbb{R}$ and $-d \leq y \leq \eta(x)$. This inequality does not hold anymore at the wave crest for the Stokes wave of greatest height, which accounts for the loss of regularity of the wave.

Another possible aspect to explore would be to remove the irrotationality constraint. An exact formula for η has still been obtained even in this case, but the results are yet to be published.

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